

The First Kalman Filter to Fly:

a MATLAB reconstruction of Apollo-era circumlunar navigation

A learning project on the linearized Kalman filter,
built from the original NASA documents

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Abstract

In 1960 a small research branch at NASA Ames set out to answer a question nobody had answered before: could a spacecraft, using nothing but a crew member's sextant and a small on-board computer, find its own way to the Moon and back? The tool they discovered—and had to reshape—became the first practical application of the Kalman filter, and eventually the navigation core of Apollo. This report retraces their path with the original documents open on the desk: the equations of motion, the optical measurement geometry, and the filter equations are taken from the 1962 technical report in which the scheme was first published, the noise levels and error budgets from the Apollo-era analyses, and the augmented error states from what Apollo 13 actually experienced in flight. Everything is reimplemented in two plain MATLAB files: a genuine ballistic free-return figure-8 trajectory is targeted in the full Earth–Moon–Sun field, star-horizon style angle measurements are generated with realistic biases, and the linearized Kalman filter is flown over the complete 5.7-day mission—once linearized about the pre-computed nominal, exactly as first formulated, and once re-linearized about the current estimate, the modification that history now calls the extended Kalman filter. The uncorrected trajectory misses Earth entry by 31,000 km; the estimate-linearized filter tracks it to about a kilometre. The gap between those two numbers is, in miniature, the story of why this filter changed spaceflight.

1 Why go back to 1960

This project began as an extension of a Kalman-filtering specialization: the course notes¹ develop the linear Kalman filter carefully and abstractly, and at some point one wants to see the machinery earn its keep on the problem it was actually invented for. For the Kalman filter, that problem is unusually well documented: the first practical application of Kalman's 1960 paper was cislunar navigation, studied at NASA Ames Research Center from 1959 onward, published in 1962, and flown—in descendant form—on every Apollo mission.

The premise of this project is therefore simple. Take the original documents. Read them the way their authors wrote them. Rebuild the whole thing—trajectory, measurements, filter—in modern MATLAB, keeping their equations, their constants, and their assumptions wherever possible, and recording every place where a simplification is made. Then fly the full mission

¹G. L. Plett, *Applied Kalman Filtering* specialization, University of Colorado Colorado Springs: Course 1, “Kalman Filter Boot Camp,” and Course 2, “Linear Kalman Filter Deep Dive.” The courses build the discrete-time predict/correct machinery — $\hat{x}_k^-$, $\hat{x}_k^+$, the innovation, the optimal gain — from sequential probabilistic inference. This project borrows their notation and their emphasis on numerically robust implementations, and applies both to the filter's first real problem.

and watch what they watched, sixty-five years later, on a machine some hundred million times faster than theirs.

An earlier version of this simulation existed, and it is worth recording why it was discarded: its “trajectory” was a straight-line interpolation between Earth and Moon, and its “truth” was defined as that same line plus noise. A filter tested against a truth that is glued to the reference by construction can only ever appear to work. The rebuild below makes the opposite choice everywhere: the reference trajectory is a genuine ballistic solution of the equations of motion, the truth is propagated independently with errors the filter is never told about, and every claimed success has to be earned against that separation.

2 Eight people and an IBM 704

The story is told first-hand in a NASA memoir written a quarter century later [2]. *Discovery of the Kalman Filter as a Practical Tool for Aerospace and Industry* (NASA TM-86847, 1985) is McGee and Schmidt’s own account of how the filter went from a fresh journal paper to flight software; it was written to record “the many problems that had to be overcome, as well as the extremely fortunate sequence of events” behind the discovery. This project takes from it the historical sequence recounted in this section, and two technical facts that shaped the implementation: the original measurement set (“elevation, azimuth, and subtended angles of Earth and Moon”), and the reason the extended Kalman filter exists at all.

In the fall of 1959—before Apollo was a national program—the Dynamics Analysis Branch at Ames chose midcourse navigation and guidance for a manned circumlunar mission as its research problem. The branch had eight analytically oriented researchers, an IBM 704, and, in their own words, “nothing in the way of analytical software tools.” They wrote their own matrix subroutines in Fortran and then rewrote them with single indexing when the double-indexed versions proved so slow that “a circumlunar trajectory with a Kalman filter would have taken hours of computer time.” The Jet Propulsion Laboratory supplied an ephemeris tape of Sun and Moon positions; by mid-1960 the branch could compute free-return trajectories to the Moon.

Navigation was the part that would not yield. The plan had always been that the crew would measure angles—the direction and apparent size of the Earth and Moon—through an inertially referenced optical instrument, and that a computer on board would turn those sightings into position and velocity. JPL’s iterative weighted least-squares machinery was too heavy for any conceivable on-board computer of the era. Fitting polynomials to angle histories was light enough but threw away accuracy. Wiener filter theory, which some of the staff knew well from missile guidance work, could not digest the nonlinearity of the problem or the irregular, discrete timing of a human navigator’s sightings. “Obviously,” the memoir says, “a new approach was needed.”

Then, in the fall of 1960, Rudolf Kalman visited Ames. He and Schmidt had known each other for years; Kalman, unaware of the lunar work, came to discuss topics of mutual interest and presented his new paper on discrete linear filtering [6]. The sequential structure “hit a responsive chord,” but the paper was hard going—the state-space language it used was new to practicing engineers—and it took a second meeting, at Kalman’s institute, before the shape of the solution came clear to Schmidt: the branch’s linear perturbation methods could manufacture exactly the linear system Kalman’s theory required. Linearize the nonlinear problem about a reference trajectory, and filter the deviations.

Two modifications had to be invented before the idea could fly, and both are now so standard that it takes effort to see them as inventions. First, Kalman’s formulation assumed regularly spaced measurements; a crew taking sextant sightings does not work that way. The Ames group decomposed the algorithm into separate time-update and measurement-update steps

so that observations could be processed whenever they happened to arrive. “Looking back, the decomposition seems almost trivial; at the time, however, it was a major and critical step.” Second—and this one arrived by accident—a simulation was once mis-configured so that the true trajectory stayed in Earth orbit while the estimate believed it was on a lunar transfer. A filter linearized about a reference would have been helpless; their implementation, re-linearized about the current *estimate*, overshoot, recovered, and converged. That modification “has come to be known as the extended Kalman filter.”

The first full run of the Kalman-filter simulation took “a little over an hour of execution time” on the IBM 704 and produced disappointing results — traced to an error in how a subroutine was called. “On the second run everything worked fine!” By early 1961 the branch had shown that on-board optical measurements plus knowledge of the equations of motion could navigate a circumlunar mission with accuracy comparable to weighted least squares, at a fraction of the memory and compute. In September 1961 Richard Battin of the MIT Instrumentation Laboratory—which would build Apollo’s guidance system—published his own optimizing navigation procedure and credited Schmidt for directing his attention to Kalman’s work. Battin’s colleague Potter later devised the square-root formulation that made the filter numerically safe on the small word length of the flight computer. A direct line runs from that fall 1960 visit to the software that navigated Apollo.

3 The report on the desk: NASA TR R-135

The formal publication of the Ames work is the document this whole project is built around [1]. *Application of Statistical Filter Theory to the Optimal Estimation of Position and Velocity on Board a Circumlunar Vehicle* (NASA TR R-135, 1962), by Smith, Schmidt and McGee, was written to put the new trajectory-estimation scheme on record: the problem formulation, the optimal filter theory, the implementation recipe for an on-board computer, and a simulation study. From it this project takes essentially everything: the equations of motion and physical constants (its Appendix A), the linear perturbation dynamics (Appendix B), the optical observables and their partial derivatives (Appendix C), the filter equations themselves (its eqns. (12)–(16)), the injection covariance and observation schedule of its simulation study, and the bias-sensitivity result that motivates one of the augmented states below.

R-135 defines the midcourse phase as “all of the flight after boost and before re-entry,” and states the reason navigation matters in one sentence that the rest of this report will keep vindicating: small errors at injection “produce such large errors later along the trajectory (for instance, near the moon) that guidance is generally necessary.”

3.1 The equations of motion, as they wrote them

Appendix A of R-135 gives the vehicle’s acceleration in a nonrotating geocentric Cartesian frame: an oblate Earth (second harmonic only), plus the Moon and Sun as spherical bodies whose attraction appears in differential form—their pull on the vehicle minus their pull on the Earth-centered frame itself. With $r = |\mathbf{r}|$, $\mathbf{r}_m$ and $\mathbf{r}_s$ the geocentric Moon and Sun positions, $\Delta_m = |\mathbf{r} - \mathbf{r}_m|$, $\Delta_s = |\mathbf{r} - \mathbf{r}_s|$, the X -component of R-135’s eqns. (A1)–(A3) reads

$$\ddot{X} = -\frac{\mu_e X}{r^3} \left[1 + J \left(\frac{a}{r} \right)^2 \left(1 - \frac{5Z^2}{r^2} \right) \right] - \mu_m \left(\frac{X - X_m}{\Delta_m^3} + \frac{X_m}{r_m^3} \right) - \mu_s \left(\frac{X - X_s}{\Delta_s^3} + \frac{X_s}{r_s^3} \right), \quad (1)$$

with Y analogous and the Z -component carrying $(3 - 5Z^2/r^2)$ in the oblateness bracket. The constants are taken verbatim from the same page: $\mu_e = 3.986135 \times 10^5$, $\mu_m = 4.89820 \times 10^3$,

$\mu_s = 1.3253 \times 10^{11}$ (all km^3/s^2), Earth equatorial radius $a = 6378.26$ km, and $J = 1.6246 \times 10^{-3}$ (R-135’s second-harmonic coefficient, $\frac{3}{2}J_2$ in modern notation).

Two simplifications are made here, and recorded. R-135’s simulation used a stored JPL ephemeris; this project puts the Moon on a circular orbit of radius 384,400 km (sidereal period 27.321661 d) and the Sun on a geocentric circle of one astronomical unit, both in the XY -plane, which is taken to be the Moon’s orbital plane rather than the equator. Nothing in the filter depends on these choices; they buy a clean, reproducible two-file implementation.

Propagation is fixed-step fourth-order Runge–Kutta on the 60-second filter grid, with each step subdivided (up to $10\times$) whenever the vehicle is within 40,000 km of either body, so that the perigee and perilune passages—where the dynamical time scale collapses—are integrated as carefully as the cruise. The 704 generation solved exactly this problem with a variable-step routine whose ability “to be stopped and restarted at any arbitrary time” the memoir singles out as critical; the proximity substepping here is the same idea in fixed-grid clothing.

4 A trajectory that comes home by itself

R-135 studied the first hours after translunar injection of a circumlunar trajectory. This project flies the whole thing, so the first task is the one the Ames group solved in mid-1960 with JPL’s help: find a genuine *free-return* trajectory—a ballistic arc that leaves a 185-km parking orbit, loops behind the Moon, and falls back to an Earth re-entry with no propulsive maneuver anywhere. The figure-8 shape of that solution is Apollo iconography, and its safety logic is why Apollo 13’s crew came home.

The targeting problem has two unknowns and two conditions. The unknowns are the injection speed v_0 and the injection position angle θ_v (injection is tangential and prograde from the parking orbit; the Moon’s initial phase is fixed). The conditions chosen here are a perilune radius of $R_M + 111$ km—an Apollo-11-class pericynthion altitude—and a return vacuum perigee of $R_0 + 40$ km, safely below the 400,000-ft entry interface so that the return arc is a re-entry, not a flyby. (The entry-interface definition is taken from the Bellcomm error analysis introduced in the next section.)

Two properties of the problem dictate the solver. First, the corridor is brutally stiff: near the solution, one metre per second of injection speed moves the perilune by thousands of kilometres. Second, the two controls separate naturally: v_0 predominantly sets the perilune height, while θ_v shapes where the post-flyby arc falls. A raw two-dimensional Newton iteration wanders in such a valley, and a coarse scan cannot land inside a corridor a metre-per-second wide; what works is a nested pair of one-dimensional secant iterations—inner loop on v_0 to pin the perilune, outer loop on θ_v to pin the return perigee—warm-started from a grid scan that also checks the *direction* of the lunar pass. Only a retrograde pass (negative z angular momentum about the Moon) bends the trajectory back toward Earth; the prograde branch slings the vehicle outward, and rejecting it in the scan is what anchors the solver on the figure-8.

Converged to sub-kilometre miss on both targets, the solution is: injection at $v_0 = 10.964236$ km/s and $\theta_v = -132.696^\circ$, perilune altitude 111.0 km at $t = 68.58$ h, return vacuum perigee 40.0 km at $t = 137.31$ h, and arrival at the entry interface after 137.28 h—5.72 days from injection to entry, with the Moon-corotating view of Fig. 1 showing the classic figure-8.

It is worth pausing on what this one step cost in 1961. The transition matrices used for filter propagation and midcourse targeting had to come from integrating “an additional set of 18 complicated, second-order differential equations” alongside the trajectory; the software went six months behind schedule; matrix elements were punched on cards to be read into the

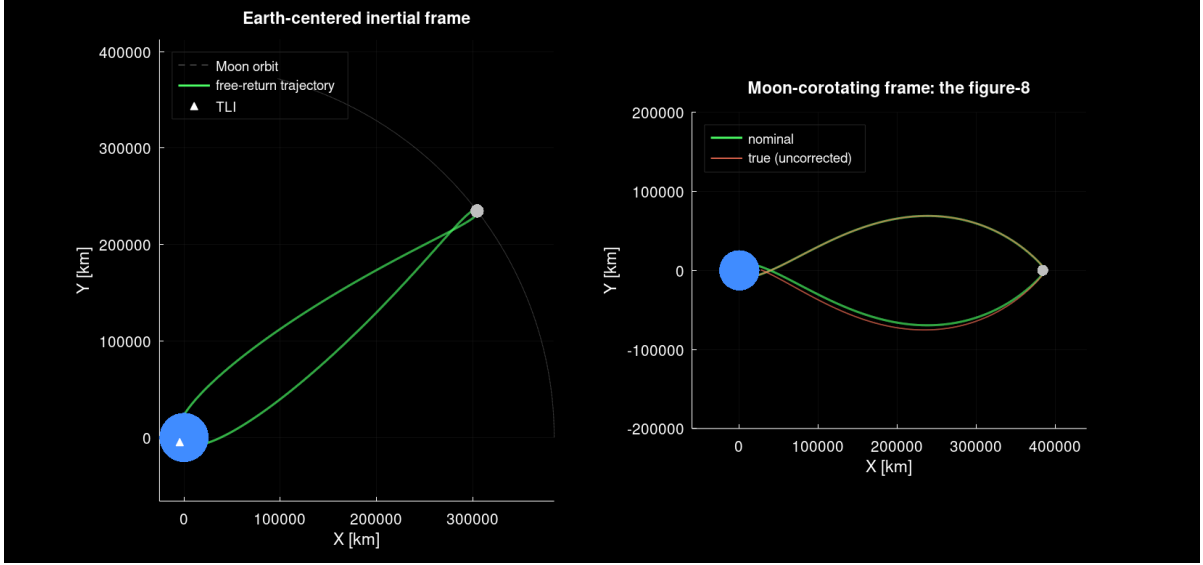


Figure 1: The targeted ballistic free-return trajectory. Left: Earth-centered inertial frame; the Moon (grey, drawn at its perilune-time position) moves along the dashed orbit while the spacecraft flies out on one arc and returns on another. Right: the same trajectory in the Moon-corotating frame—the figure-8. The true (uncorrected, dispersed) trajectory is overlaid in red; its separation from the nominal after the lunar pass is the error amplification this whole subject exists to fight.

simulation [2]. The equivalent operation below is a call to `expm`.

5 What the navigator saw

5.1 Three angles to the Earth, three to the Moon

Appendix C of R-135 defines the observables: the direction of the vehicle–Earth line, expressed as two angles, and the apparent size of the Earth, expressed as the half subtended angle. With (X, Y, Z) the vehicle’s geocentric position and R its magnitude, R-135’s eqns. (C1) are

$$\alpha_e = -\sin^{-1} \frac{Z}{R}, \quad \beta_e = \text{atan2}(-Y, -X), \quad \gamma_e = \sin^{-1} \frac{R_0}{R}, \quad (2)$$

where $R_0 = 6371$ km is the horizon reference radius and the `atan2` form of β_e reproduces exactly the partial derivatives R-135 tabulates. Those partials (R-135, App. C table) populate the measurement matrix; for one angle triple,

$$H_1 = \begin{bmatrix} \frac{XZ}{R^2\rho} & \frac{YZ}{R^2\rho} & \frac{Z^2 - R^2}{R^2\rho} \\ -\frac{Y}{\rho^2} & \frac{X}{\rho^2} & 0 \\ -\frac{R_0 X}{R^2 s} & -\frac{R_0 Y}{R^2 s} & -\frac{R_0 Z}{R^2 s} \end{bmatrix}, \quad \rho = \sqrt{X^2 + Y^2}, \quad s = \sqrt{R^2 - R_0^2}, \quad (3)$$

and, because angles depend on position only, the full measurement matrix is $H = [H_1 \ 0]$ (R-135, p. 13). The memoir confirms that the intended instrument measured “the elevation, azimuth, and subtended angles of Earth *and* Moon” [2]; accordingly, this simulation forms the same triple relative to the Moon (with $R_M = 1737.4$ km as the reference radius) whenever the vehicle is inside the lunar sphere of influence, $\approx 66,200$ km. Two visibility effects are modeled because the geometry imposes them: the first sighting is taken half an hour after injection,

matching the schedule of R-135’s own simulation, and Earth sightings are dropped whenever the Moon occults the line of sight—which happens, satisfyingly, exactly around perilune.

5.2 How good was a sextant, and what was it biased by

Numbers for the measurement noise come from the Apollo program’s own error analysis [3]. The Bellcomm memorandum *Summary of Apollo Guidance and Navigation Error Analysis* (TR-66-310-4, 1966) condensed a large TRW-supported Monte Carlo study of the full lunar-orbit-rendezvous mission, run separately for ground-radar and on-board-optical navigation; its purpose was to verify that mission strategy was consistent with hardware performance. This project takes from it the nominal sextant accuracy—**10 arc-seconds** (1σ), with a degraded 30-arc-second case studied—the definition of the entry interface at 400,000 ft, and a sobering headline it is only honest to repeat: ground-based radar navigation was found “clearly superior” to the on-board optical system. The on-board system earned its place as the self-sufficient backup—which is precisely the Apollo 13 scenario. All six angles here carry independent Gaussian noise with $\sigma = 10''$, R-135’s own assumption of independent, identically distributed angle errors (it studied σ_n from $5''$ to $200''$).

Real sightings were not merely noisy; they were biased, and the flight data says by how much. The Apollo 13 guidance report [4] (*Apollo 13 Mission Report, Supplement 1: Guidance, Navigation and Control Systems Performance Analysis*, TRW for NASA MSC, 1970) is the engineering post-mortem of CSM-109’s GN&C hardware; it was written to reconstruct instrument performance through the aborted mission. Two of its findings are built into the truth model here. First, processing of the translunar P23 star-horizon sightings showed “the actual horizon altitude was about 18 kilometers” while “the apparent altitude (due to small sighting errors) was about 10 kilometers” (its sec. 3.2): the subtended-Earth-angle measurement carried a systematic horizon-altitude error of order 8 km. Second, immediately after the oxygen-tank incident the accelerometers registered low-level outputs “which resembled the effect of venting”—an effect ultimately attributed to a power-supply transient, but venting-type disturbance accelerations on a spacecraft that is supposed to be in free fall are exactly the kind of unmodeled dynamics a navigation filter must survive (its sec. 3.1.4).

R-135 anticipated both, and told us what to do about them. Its bias study (p. 21) added a constant $5''$ bias to one angle at a time and found the system “quite a bit more sensitive to bias errors on γ_e ”—the subtended angle—“than on α_e and β_e ”; and it noted that “such bias errors can be treated as additional state variables and estimated along with the others. The theory presented is sufficiently general to allow this.” Done:

- the **truth** generates Earth- γ measurements with an effective horizon radius $R_0 + b_h$, $b_h = 8$ km (the Apollo 13 number), and flies with a constant unmodeled acceleration $\mathbf{a}_u = [-3, 6, -2] \times 10^{-9}$ km/s² (≈ 0.7 micro- g , a venting-class disturbance);
- the **filter** carries both as augmented states: a 10-state vector $x = [\delta\mathbf{r}; \delta\mathbf{v}; \mathbf{a}_u; b_h]$, with $\partial\gamma_e/\partial b_h = 1/\sqrt{R^2 - (R_0 + b_h)^2}$ appended to (3).

5.3 The truth the filter never sees

The true trajectory starts from the nominal injection state plus an injection error, and here the reconstruction borrows R-135’s own random draw: the report lists the sampled injection error used in its simulation study (p. 19), $[0.495, -0.886, -1.001]$ km and $[0.281, 1.999, 0.194]$ m/s. Applied at full size to a mission with no midcourse corrections, that error is spectacular—R-135 itself quotes a 4528-km miss at perilune for its reference trajectory, and on this one the

uncorrected pass would strike the Moon. A real crew would have corrected long before; rather than model burns, this simulation scales the R-135 sample by 0.05 (a post-midcourse dispersion level, consistent in size with the metres-per-second TLI velocity errors tabulated in [4]) and orients it so the dispersion *raises* the pass. Both choices are recorded design decisions. The result: the true perilune altitude is 243.6 km against the nominal 111 km, a 344-km deviation at the Moon that the lunar flyby then amplifies, ballistically and mercilessly, to **31,312 km** by the time the nominal trajectory reaches entry. R-135’s one-sentence justification for midcourse guidance, quantified.

6 The filter, in their equations

6.1 Perturbation dynamics

Appendix B of R-135 linearizes (1) about the reference: the deviation state obeys $\dot{x} = F(t)x$ with

$$F = \begin{bmatrix} 0_{3 \times 3} & I_3 \\ G & 0_{3 \times 3} \end{bmatrix}, \quad G = \frac{\partial \mathbf{a}}{\partial \mathbf{r}} = \sum_{b \in \{e, m\}} \mu_b \left(\frac{3 \mathbf{d}_b \mathbf{d}_b^\top}{d_b^5} - \frac{I_3}{d_b^3} \right), \quad \mathbf{d}_b = \mathbf{r} - \mathbf{r}_b, \quad (4)$$

which is R-135’s eqn. (B7) with the gravity gradient evaluated analytically for the point-mass Earth and Moon (the J_2 and solar contributions to G are second order here and are retained only in the trajectory propagation itself). With the augmented states— $\mathbf{a}_u$ entering $\dot{\mathbf{v}}$ directly, b_h constant—the 10-state system matrix and its discretization over one 60-s grid step are

$$A = \begin{bmatrix} 0 & I_3 & 0 & 0 \\ G & 0 & I_3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \Phi_k = e^{A(t_k) \Delta t}. \quad (5)$$

R-135’s Appendix E describes how to obtain Φ on the 704: place a unit initial condition on one state, integrate the perturbation equations, read off a column, repeat six times. MATLAB’s `expm` does this in one line, roughly eight thousand times per filter run, without comment. Process noise is standard white-acceleration discretization ($Q_{rr} = q\Delta t^3/3$, $Q_{rv} = q\Delta t^2/2$, $Q_{vv} = q\Delta t$, with $q = (4 \times 10^{-11} \text{ km/s}^2)^2 \text{ s}$) plus a slow random walk on $\mathbf{a}_u$; the disturbance itself is a state, so Q only needs to cover linearization residue and integration error.

6.2 The estimation equations

The heart of R-135 is its eqns. (12)–(16), which any student of the modern literature will recognize on sight:

$$\hat{x}(t_k) = \Phi \hat{x}(t_{k-1}) + K(t_k)[y(t_k) - H \Phi \hat{x}(t_{k-1})], \quad (\text{R-135: 12})$$

$$K(t_k) = P(t_k)H^\top [HP(t_k)H^\top + Q_n]^{-1}, \quad (\text{R-135: 13})$$

$$P(t_{k+1}) = \Phi[P(t_k) - K(t_k)HP(t_k)]\Phi^\top + Q^*, \quad (\text{R-135: 14–16})$$

i.e. exactly the predict/correct pair of the course notes, with $\hat{x}_k^- = \Phi \hat{x}_{k-1}^+$, innovation $y_k - H \hat{x}_k^-$, and gain $K_k = P_k^- H^\top (HP_k^- H^\top + R)^{-1}$; only the symbols have aged. Two implementation choices in this project are themselves period-faithful. First, the measurement update is processed *one scalar angle at a time*: R-135 points out (p. 23) that with independent errors each component can be treated as its own observation, so that “the matrix to be inverted is in this case 1×1 , and the inversion is trivial so that the ultimate in calculation simplicity is realized.” Second, the covariance update uses the Joseph form, $P^+ = (I - KH)P^-(I - KH)^\top + KRK^\top$, with explicit

Table 1: Recorded design decisions and their sources.

Choice	Value	Source / rationale
Equations of motion	Earth(J_2)+Moon+Sun, differential form	R-135 App. A, eqns. (A1)–(A3)
Constants $\mu_e, \mu_m, \mu_s, a, J$	R-135 values, km–s units	R-135 App. A
Ephemerides	circular, coplanar Moon/Sun	simplification (R-135 used JPL tape)
Reference trajectory	ballistic free return, figure-8	Apollo free-return doctrine
perilune target	$R_M + 111$ km	Apollo-11-class pericynthion
return perigee target	$R_0 + 40$ km < entry interface	entry at 400 kft [3]
Grid / integrator	60 s, RK4 + proximity substeps	filter grid; perilune accuracy
Observables	α, β, γ of Earth; of Moon in SOI	R-135 App. C; [2]
Angle noise	$\sigma = 10''$, i.i.d., all angles	[3]; R-135 p. 17
Schedule	every 15 min from $t=30$ min	R-135 p. 16 schedule, extended
Occultation	Earth angles dropped behind Moon	geometry
Injection error (truth)	$-0.05 \times$ R-135 sampled error	R-135 p. 19; scale/sign recorded
Disturbance accel. (truth)	$[-3, 6, -2] \times 10^{-9}$ km/s ²	venting-class; [4] §3.1.4
Horizon bias (truth)	$b_h = 8$ km on γ_e	[4] §3.2 (18 vs 10 km)
Augmented states	$\mathbf{a}_u$ (3), b_h (1)	R-135 p. 21 recommendation
P_0	1 km, 1 m/s per axis (1σ)	R-135 p. 18
Update form	scalar sequential; Joseph + symmetrize	R-135 p. 23; numerical hygiene

re-symmetrization—the modern, numerically conservative cousin of the precautions the pioneers took (the memoir notes they were saved from round-off trouble partly by the lucky choice of evaluating $(KH)P$ rather than $K(HP)$; the Apollo flight software ultimately went further, to Potter’s square-root formulation, because the flight computer’s 15-bit words demanded it [2, 5]).

The initial covariance is R-135’s: 1 km and 1 m/s per axis (1σ), diagonal, plus 10^{-8} km/s² on each disturbance axis and 10 km on the horizon bias.

6.3 One filter, two linearizations

Everything above can be linearized in two places, and the difference between them is the hinge of the whole story.

LKF. In the filter exactly as formulated in R-135, F , Φ and H are evaluated on the *pre-computed nominal trajectory*, and the state is the deviation from it. This is the textbook linearized Kalman filter of Course 2 of the specialization.

EKF variant. R-135, p. 15, already recommends better: computing H and Φ in the vehicle “makes it possible to linearize around the estimated rather than the reference trajectory. This is clearly the correct procedure since... the estimate is on the average closer to the true state than is the reference,” and it notes that with this procedure even “very large deviations from the reference... would not jeopardize the accuracy of the trajectory determination scheme.” The 1985 memoir names what this became. In this variant the total estimate is propagated through the full nonlinear dynamics (including the estimated $\mathbf{a}_u$), and F , H are evaluated at the estimate.

Both filters run on identical measurements, identical tuning, identical initial covariance. Nothing differs but the linearization point.

7 Flying the mission

Table 1 collects the configuration; 548 sighting epochs produce 1986 scalar angle measurements over the 137.3-hour mission (six angles are lost to lunar occultation of the Earth, right at

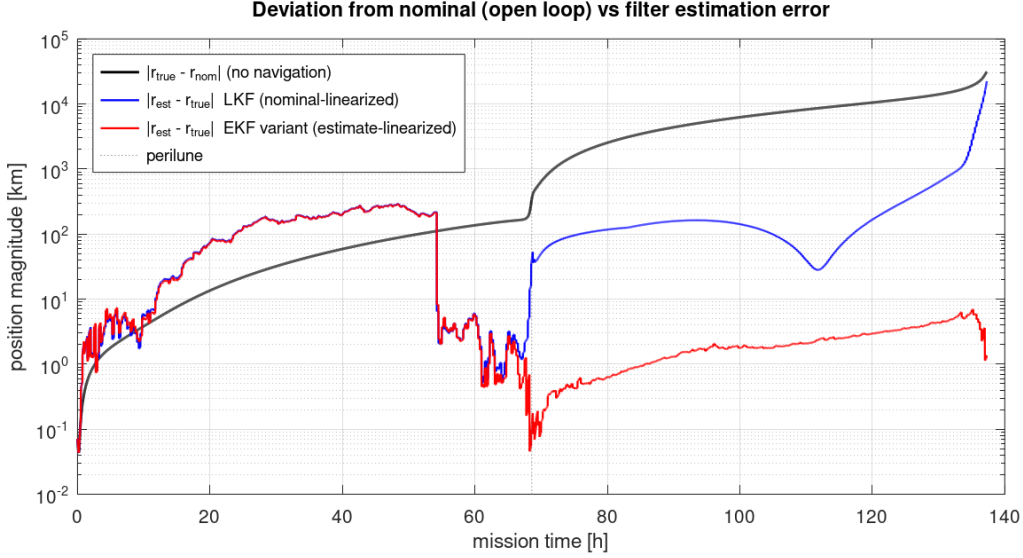


Figure 2: Open-loop error amplification versus filtered knowledge. Black: distance between the true and nominal trajectories if no navigation is performed—344 km at perilune (dotted line), 31,312 km at entry. Blue: LKF estimation error. Red: the estimate-linearized (EKF) variant. The mid-cruise plateau, the collapse at ~ 55 h, and the post-flyby split between the two filters are each discussed in the text.

perilune). Figure 2 is the mission in one picture, and repays a slow reading.

Hours 0–10. Both filters converge from the injection uncertainty to kilometre-class errors. Near the Earth the subtended angle γ_e is large and rich in range information; navigation is easy, and R-135’s own 2.5-hour simulation lived entirely in this regime.

Hours 10–55: range-blindness. As the Earth shrinks behind the spacecraft both filters’ errors grow to ~ 300 km, riding their own 3σ envelopes (Fig. 3). This is not a bug; it is geometry, and R-135 diagnosed it in 1962: the estimation-error ellipsoid’s “major axis... tends to be oriented along the vehicle-earth line,” because direction angles pin the transverse position while the range must come from γ_e , whose information content degrades as R_0/R^2 — “the information available from this angle is relatively poor” (R-135, p. 20). Sixty-five kilo-km from home, a ten-arc-second sextant knows *where you appear against the stars*, not *how far out you are*.

Hour ~ 55 : a second lighthouse. The vehicle crosses into the lunar sphere of influence, the Moon triple switches on, and the range ambiguity dies instantly—both filters collapse to km-level in a single measurement batch, the sharp cliff in Figs. 2 and 3. Geometry gives what more staring at the Earth never could: triangulation.

Perilune, hour 68.6. The Moon occults the Earth; for a few sightings the filter lives on Moon angles alone. The covariance breathes up and back down. (The crew of Apollo 8, at this point in their flight, were reading the first Genesis verses to a third of humanity; the filter just kept squaring away residuals.)

The return leg: the two filters part company. Through the flyby the true trajectory—344 km from the nominal at closest approach—sees a measurably different Moon than the nominal did, and the deviation x the LKF must estimate grows to hundreds, then thousands of kilometres. The linearization $H(\mathbf{r}_{\text{nom}})$, $\Phi(\mathbf{r}_{\text{nom}})$ is now simply evaluated in the wrong place. The LKF holds on gamely at 100–200 km of error for most of the return coast, and then, as the geometry sharpens near Earth and the measurements become most informative, the mismatched H turns that information into poison: the LKF diverges to 22,000 km by entry (Fig. 4), its augmented states swallowing the linearization error whole (Fig. 5). The estimate-linearized variant, identical in every respect but its linearization point, glides through the same data to

a **1.4 km** final position error, its innovations white and inside $\pm 3\sigma$ from injection to entry (Fig. 6). R-135’s p. 15 recommendation, the accidental discovery of 1961, the reason “extended” precedes “Kalman filter” in every textbook: reproduced in one plot.

The parameter states tell their own quiet story (Fig. 5): the EKF identifies all three components of a $0.7 \mu g$ venting acceleration it was never told about, and pins the Earth-horizon bias at 8 km—the very discrepancy the Apollo 13 analysts dug out of the P23 sightings—within the first hours, because near Earth $\partial\gamma_e/\partial b_h$ is large exactly when γ_e is most informative.

8 Then and now

It is worth ending where the reconstruction kept ending: on the distance between what this took in 1961 and what it takes now.

The Ames feasibility studies ran on an IBM 704. The Kalman-filter program occupied about 13,000 words and needed roughly *fifteen minutes* of 704 time to process a simulated 2.5-hour flight (R-135, p. 18); the full simulation with trajectory integration took over an hour per run [2]. Transition matrices meant integrating eighteen additional second-order differential equations; matrix inversion was something you restructured your algorithm to avoid; a misplaced subroutine call cost a day. The computer that eventually flew this mathematics—the Block II Apollo Guidance Computer documented in the Draper Laboratory’s final report [5]—had 15-bit words plus parity, 36,864 words of fixed (rope) memory, 2,048 words of erasable memory, a 46.8-microsecond multiply, and weighed 70 pounds while drawing 55 watts. *MIT’s Role in Project Apollo, Volume III* (R-700 vol. III, 1972), from which those numbers come, is the final report on that machine’s design and manufacture—the five-volume set opens with Kennedy’s May 1961 commitment, and volume III is the story of fitting guidance, navigation and a Kalman-descended filter into one cubic foot of core rope. On that word length, the covariance recursion in its textbook form loses positive-definiteness; that is why Potter’s square-root filter, not the plain recursion, is what actually flew [2].

The reconstruction described in this report is two MATLAB files. The complete pipeline—corridor targeting of the free-return trajectory, independent truth propagation, 1986 sequential scalar measurement updates through two ten-state filters, all figures—runs in about two minutes of laptop time, most of it spent in the trajectory targeting. Φ_k is `expm(A*dt)`. The covariance stays symmetric because two extra characters, $(P+P')/2$, are free. The animation that would have been a plotter-and-film-camera project renders in minutes. None of this is a boast about now; it is a measurement of them. Eight people, one 704, and a visitor with a strange new paper built, in about eighteen months, the navigation architecture of crewed spaceflight—and the hard parts they solved (which linearization point; how to decompose the update; how to keep a covariance honest on short words) are still the hard parts, just wearing better tools.

9 What this project taught

Three things, in the end. First, the linearized Kalman filter is exactly as good as its reference: flawless for fifty hours, fatally wrong-footed the moment a lunar flyby amplified a 344-km deviation into thousands. Second, the fix was known to the filter’s first users before the filter had a name—linearize about the estimate—and seeing the two variants diverge on identical data is worth more than any convergence proof. Third, the historical documents are not garnish. Every constant in the truth model of this simulation—ten arc-seconds, eight kilometres of horizon, a micro- g of venting, kilometres-and-metres-per-second of injection error—came out of a report written by people who had to get this right with three astronauts attached. Reading

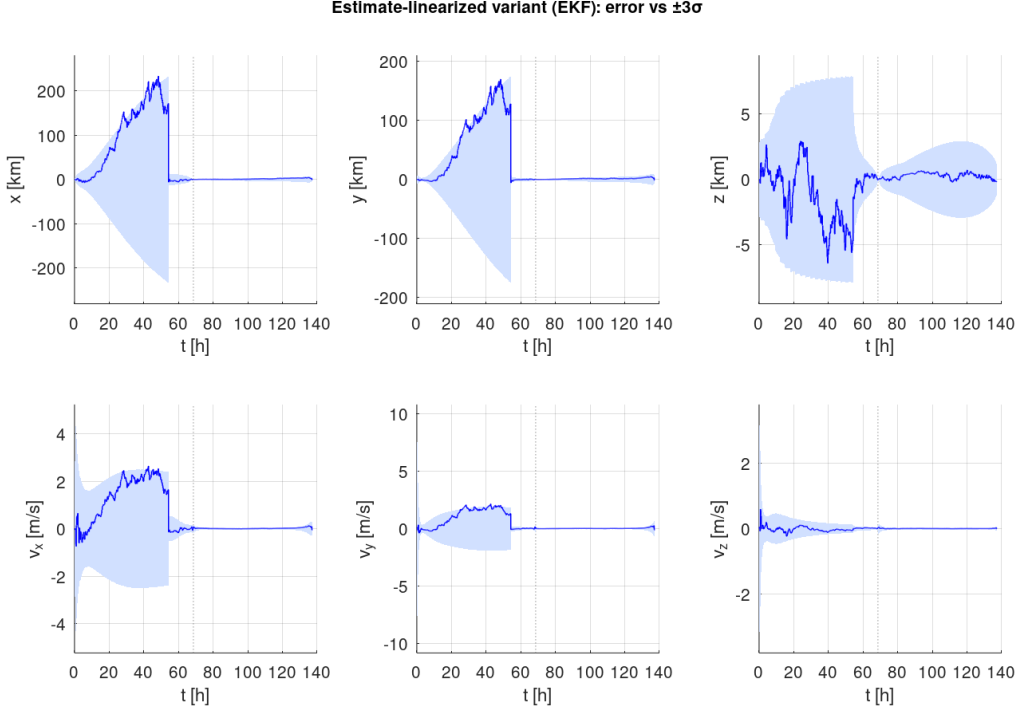


Figure 3: Estimate-linearized (EKF) variant: estimation errors against $\pm 3\sigma$. The mid-cruise growth is the range-blind regime; the collapse at ~ 55 h is the entry into the lunar sphere of influence. The filter is statistically consistent for the entire mission.

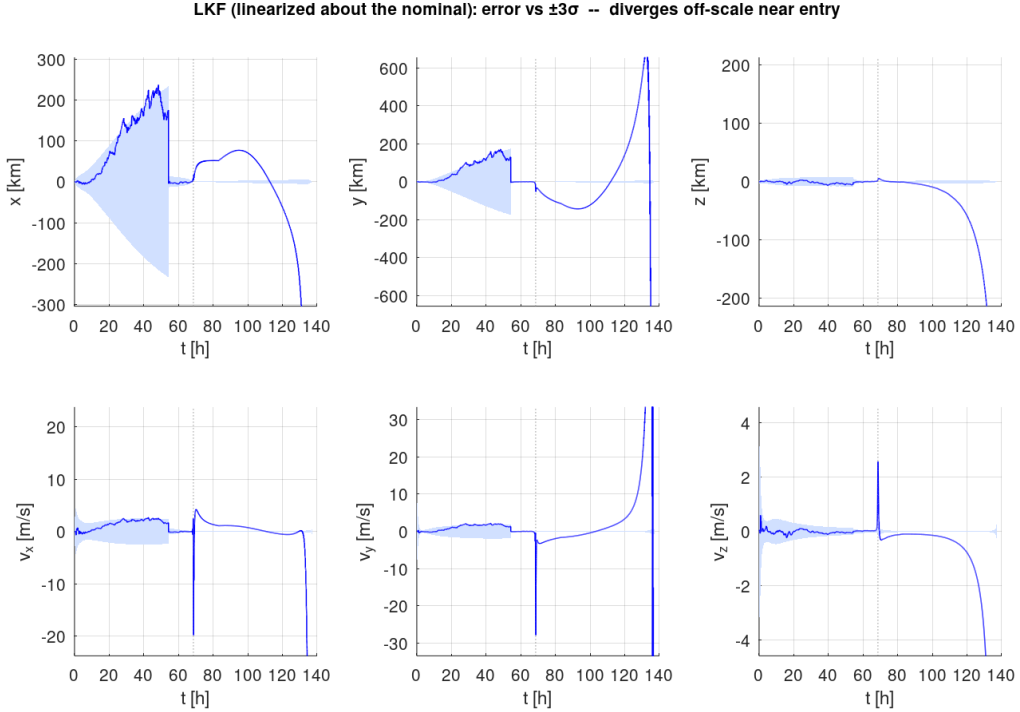


Figure 4: The LKF, linearized about the nominal, on the same data and the same axes-scale philosophy (panels clipped to the pre-breakdown scale). Identical to the EKF until the flyby; afterwards the nominal-based linearization degrades and the filter finally diverges off-scale as the most informative measurements of the mission arrive with the wrong H .

Augmented states: venting acceleration and Earth-horizon bias (LKF runs off-scale after the flyby)

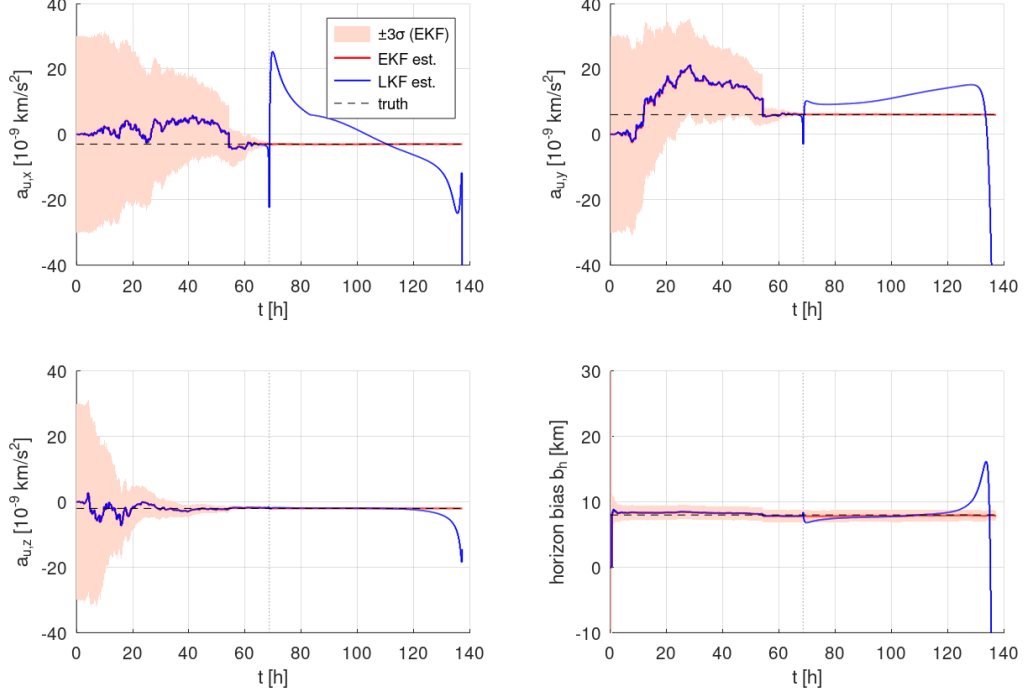


Figure 5: Augmented states. Red: EKF estimates with $\pm 3\sigma$ bands; blue: LKF; dashed: truth. The venting acceleration ($0.7 \mu g$ -class, motivated by Apollo 13’s post-incident PIPA episode) and the 8-km Earth-horizon bias (the P23 finding) are both recovered by the EKF; the LKF’s parameter states absorb its post-flyby linearization error and run off-scale.

Normalized subtended-Earth-angle innovations (EKF variant)

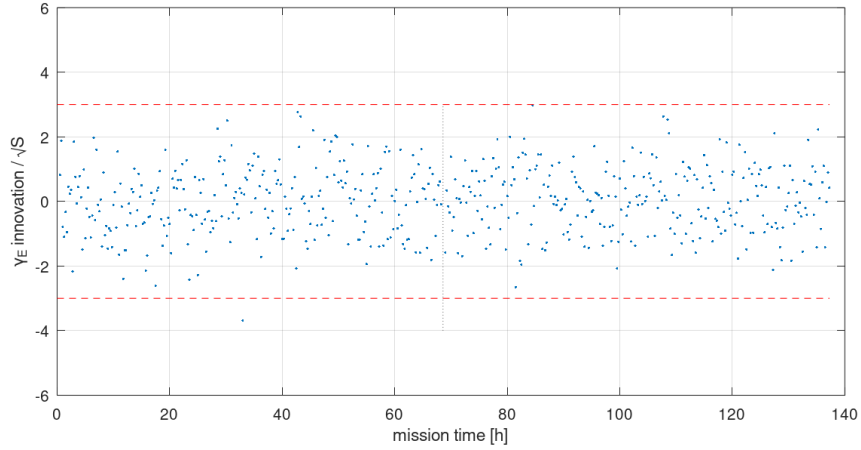


Figure 6: Normalized subtended-Earth-angle innovations of the EKF variant: white, zero-mean, inside ± 3 for the whole mission — the filter’s internal consistency check, on the very angle R-135 found most bias-sensitive.

them is the closest thing the subject has to a time machine.

References

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Companion code: `apollo_lkf_mission.m` (trajectory targeting, truth simulation, both filters, all figures) and `apollo_lkf_animation.m` (mission animation). Base MATLAB, no toolboxes; every model choice is annotated in the source with the citation it came from.